Conditional DiffTraj on Porto taxis and Turin buses

Jingting Liu

Question and data

Given a taxi's starting area, destination, travel time and distance, can DiffTraj generate journeys that resemble recorded trips? Zhu et al. tested this on Porto taxis in Appendix C of their 2023 paper. We repeated the experiment on long Porto journeys, then trained the same model on a day of bus journeys that we recorded in Turin.

The UCI Porto dataset contains one GPS trace per taxi journey, with positions recorded every 15 seconds. We kept complete traces with valid coordinates and at least 120 recorded positions. At this sampling rate, the shortest eligible trace spans 29 minutes and 45 seconds. We chose the cutoff described in Appendix B.1 of the paper. The authors report their Porto scores separately in Table 5 and do not give the exact set of Porto trips they used.

Of the 1,710,670 original records, 55,267 journeys from 439 taxis met our criteria. We split the journeys by taxi ID so that each taxi appeared in only one set. This gave 46,470 training, 5,241 development and 3,556 test journeys. We used 512 development journeys to choose the model checkpoint and the same 3,000 test journeys to score both runs.

Model and training

We interpolated each trip to 200 longitude–latitude pairs. During training, we added Gaussian noise to these coordinates and trained a temporal U-Net to predict the added noise. The network also received eight trip attributes: departure time bin, distance, duration, original point count, average displacement between observations, average speed, origin cell and destination cell. To generate a trip, the model starts with random noise and removes it over 100 steps using the supplied attributes. We then interpolated the generated coordinates back to the trip's original point count. Since the model receives the target distance, the length score measures how closely its sampled path follows that request.

We trained the released network from scratch twice, with seeds 42 and 43. Both runs used 200 epochs, 500 noise levels and a total batch size of 1,024 across four NVIDIA L20 GPUs. We normalized the inputs using training data. The authors' training script uses AdamW with a fixed learning rate. After an earlier Porto run produced unstable results, we clipped the gradient norm at 1 and used a cosine schedule to reduce the learning rate from 0.0002 to 0.00002. The lowest development noise-prediction loss occurred at epoch 193 for seed 42 and epoch 179 for seed 43. Both checkpoints were evaluated with the same requests and sampling noise.

Results against the published Porto benchmark

Density and OD errors compare point occupancy and ordered origin–destination cell frequencies on a 16×16 grid. Length error compares distributions of cumulative journey distance. All three use Jensen–Shannon divergence (JS), with zero for identical distributions. Our length histogram uses 20 training-quantile bins. Our pattern F1 measures the overlap of the 20 most frequently visited cells.

	Density JS ↓	OD JS ↓	Length JS ↓	Pattern F1 ↑
Published Table 5	.00870	.01320	.02420	.847
Long trips, seed 42	.01357	.00594	.00635	.750
Long trips, seed 43	.00944	.00698	.00395	.800

How the grid changes hotspot F1. We scored each set of 3,000 draws against identical observed journeys under three grid definitions. Each grid divides the specified longitude and latitude ranges into 16 equal intervals per axis. We count GPS points in each cell and rank the cells separately for observed and generated trips. Both hotspot sets contain 20 cells, so F1 equals the number of shared cells divided by 20.

Training coordinates used to set grid edges	Seed 42: hits / F1	Seed 43: hits / F1
Long trips, full min–max (primary)	15/20: 0.750	16/20: 0.800
All eligible lengths sampled for training, full min–max	17/20: **0.850**	16/20: 0.800
Same sample, 1st–99th percentile per axis	15/20: 0.750	16/20: 0.800

For seed 42, moving the grid boundaries increased the overlap from 15 to 17 cells. Seed 43 recovered 16 cells on all three grids. The generated journeys were identical across grid definitions for each seed. The full-range grids assign points outside the bounds to the nearest edge cell. The percentile grid counts those points separately and excludes them from the hotspot ranking. The published F1 is 0.847. The paper does not specify the Porto boundaries or how many hotspots were counted, so the closest numerical match (0.850) has no special status.

Trip length in kilometres. The observed median was 13.96 km. The generated medians were 12.39 km for seed 42 and 12.64 km for seed 43. The Wasserstein distance (W1), which measures the average gap between corresponding quantiles of two length distributions, was 2.28 km and 1.92 km, respectively. Both runs therefore produced a shorter typical trip, even though their length histograms scored well by JS divergence.

Turin: collecting vehicle positions and reconstructing trips

The Turin transport operator, Gruppo Torinese Trasporti (GTT), publishes a live snapshot of vehicle positions. To build a journey history, a Python collector on the research server requested the GTT vehicle-position feed, processed the response and waited 60 seconds before the next request. Each response was a binary GTFS Realtime Protocol Buffers message containing reports from multiple vehicles. The collector archived each distinct nonempty payload as a compressed `.pb.gz` file and decoded the position reports into SQLite.

Observation format. One SQLite row represents one vehicle fix. The fields below are listed in storage order.

Field	Meaning
`vehicle, trip, service_date, route`	Vehicle ID, scheduled trip ID, service date and route ID.
`t`	Vehicle-reported Unix timestamp in seconds. Fetch time was recorded separately for each poll.
`lat, lon`	Latitude and longitude in decimal degrees.
`speed`	Optional reported speed in m/s. Empty for all fixes on this service day.

Repeated snapshots could contain the same fix. The key (vehicle, trip, service date, vehicle timestamp) prevented double counting. From 02:47 on 12 November to 02:46 on 13 November 2026 (Turin time), 1,393 polls yielded 354,114 distinct fixes from 643 vehicles on 94 routes. Of those polls, 110 returned no vehicles. The median interval between successive observations was 63 seconds.

Journey reconstruction. Fixes were grouped by vehicle, service date and trip, then sorted by vehicle timestamp. GTT's static GTFS timetable supplied the active service, route type and scheduled terminals. A new segment began at a gap over 180 seconds, a nonpositive time interval or implied speed over 35 m/s. Retained segments contained at least ten fixes, ten minutes and 500 m of travel. Each endpoint had to lie within 200 m of its scheduled terminal and within 15 minutes of its scheduled time; observed duration had to cover at least 70% of the scheduled duration. Checking bus routes against the operator's fleet catalogue yielded 3,262 bus journey candidates.

Analysis and model formats. The cleaned gzip-compressed CSV retained one row per fix, with segment ID, vehicle mode, fleet and terminal-check fields. Its interval speed is the great-circle distance between successive fixes divided by their actual timestamp difference. For training, each segment became an ordered longitude–latitude sequence in a compressed NumPy archive (`.npz`), with vehicle ID, start time, elapsed duration and original point count. A median bus journey contained 46 reported positions. Interpolation to 200 model positions was applied only after reconstruction.

Turin buses: training and evaluation

We kept each Turin vehicle in a single split, giving 2,555 training, 366 development and 341 test journeys. We trained new weights on the bus data using the same network and eight trip attributes as in Porto. Duration came from the vehicle timestamps and departure time was expressed in Turin local time. All journeys were bus trips, so the model received no vehicle-class label.

With a batch of 1,024, the Turin training set filled only three batches per epoch. Over 200 epochs the model therefore updated its weights 600 times, compared with 9,200 updates in Porto. We repeated training with a batch of 56, which gave 46 updates per epoch and 9,200 in total. We kept the journeys, epoch count and evaluation requests fixed. Changing batch size affects both the number of updates and the gradient used at each update.

Turin, 341 reserved requests	Density JS	OD JS	Length JS	Hotspot F1
Batch 1,024, seed 42	.0607	.6513	.1387	.85
Batch 1,024, seed 43	.4931	.6889	.3718	.15
Batch 56, seed 42	.0120	.3187	.0515	.95
Batch 56, seed 43	.0171	.3375	.0611	.95
Observed development vs. reserved	.0587	.3136	.0895	.75

Both smaller-batch models recovered 19 of the 20 observed hotspots. Generated median lengths were 10.01 and 10.34 km against 11.13 km observed. At seed 43 the fraction of generated points outside the training bounds fell from 70.41% to 0.36%. Length W1 fell from 8.86 to 0.54 km at seed 42 and from 95.80 to 0.51 km at seed 43.

For an equal-size comparison, 50 subsets of 341 Porto requests gave median length JS of .0157 and .0121 and OD JS of .0124 and .0126 for seeds 42 and 43. Turin showed larger errors on both measures. The observed Turin OD divergence of .3136 between disjoint vehicle groups illustrates the variation in route coverage within this service day. These results cover one service day. Testing on another date would show whether the same pattern holds when the vehicles and services in operation change.

Sources

Zhu et al., DiffTraj, NeurIPS 2023, especially Appendices B.1, B.2 and C.1.
UCI Porto taxi trajectory corpus.
Author implementation and local evaluation records.

Turin observations: GTT vehicle positions and static GTFS. Turin paired evaluation records.

What the Turin scores measure

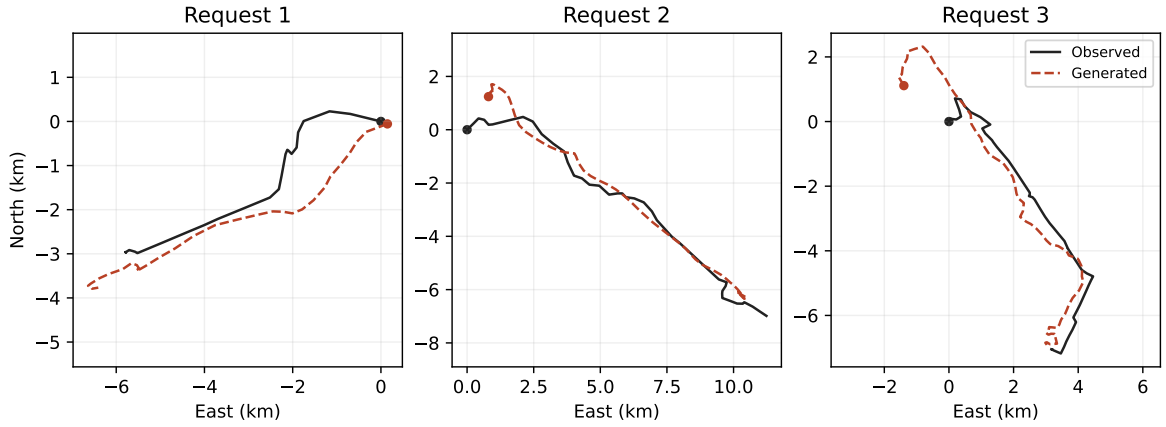


Figure 1. The first three reserved requests in the fixed evaluation order. Observed and generated positions (batch 56, seed 42) share local axes relative to the observed start. Dots mark departures. Connecting consecutive GPS fixes shows the sampled path at the reporting cadence.

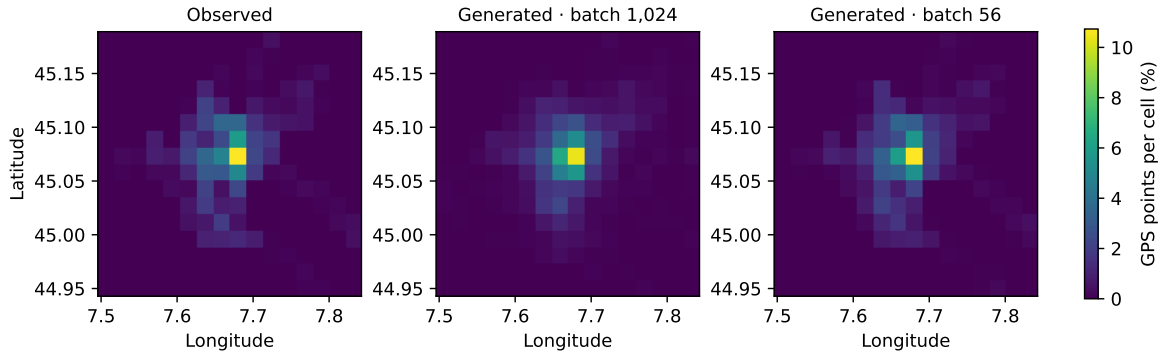


Figure 2. Share of GPS points in each grid cell for the 341 test journeys. The observed trips and both sets of seed-42 samples use the same grid and colour scale. A bright cell marks a frequently visited area; it can contain points from journeys with different origins and destinations.

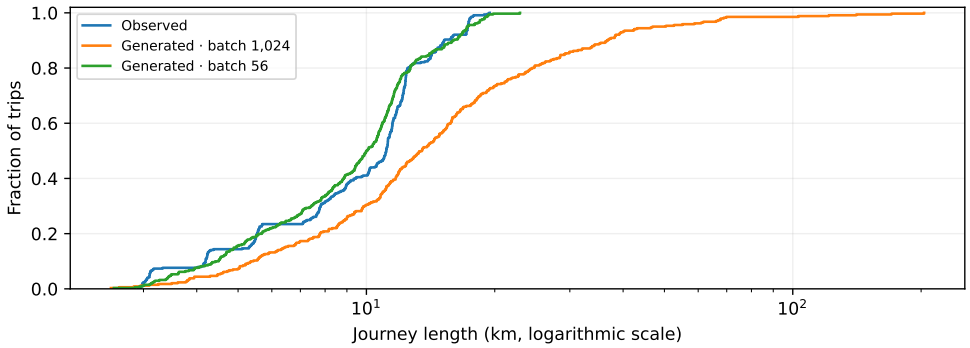


Figure 3. Empirical journey-length distributions for the same requests and seed. The logarithmic distance axis includes the upper tail. At batch 56 the generated median is closer to the observed median.